

Exp.No 4: OFDMA Resource Block Allocation

OBJECTIVE 1 :

To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.

CODE:

```
clc; clear; close all;

% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB

% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);

% Users
users = 1:6;

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) +
1; utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')
disp(total_RB)

disp('RB Allocation per User:')
disp(alloc)

figure

% ----- Graph 1: RB Allocation per User -----

subplot(2,1,1)

stem(users, alloc, 'filled', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on
```

```
% ----- Graph 2: RB Utilization -----

subplot(2,1,2)

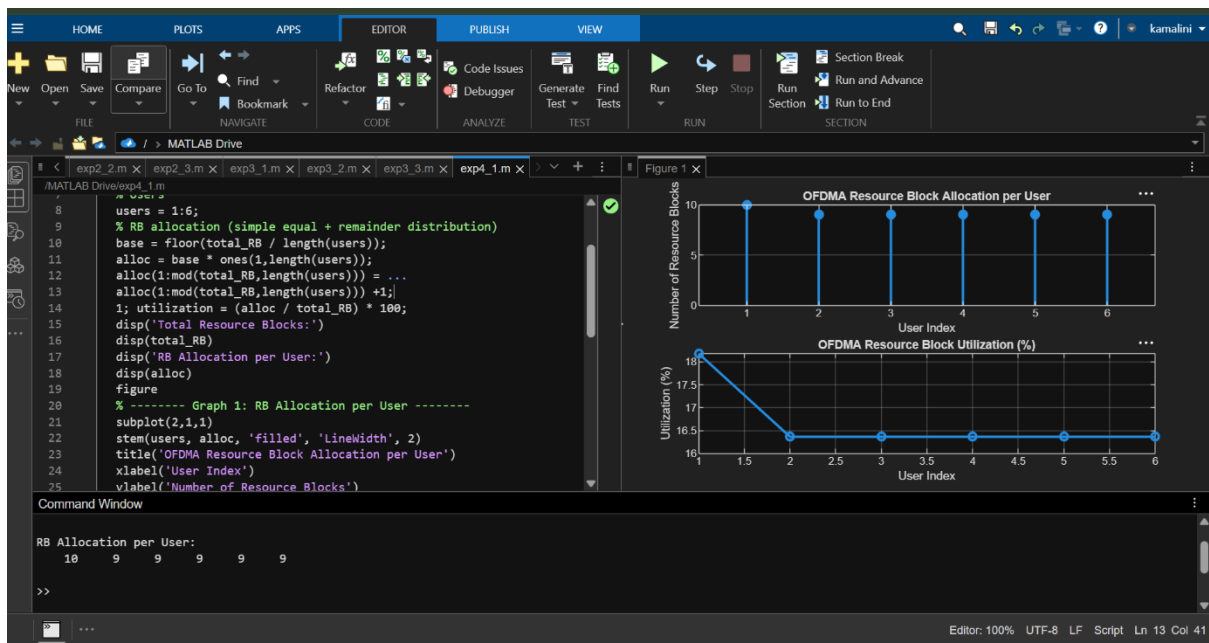
plot(users, utilization, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on
```



OBJECTIVE 2 :

To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.

CODE:

```
clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
```

```

alloc = [12 10 9 8 11];

% Resource Block Utilization (%)

util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')

disp(util)

figure

% ----- Graph 1: RB Allocation -----

subplot(2,1,1)

plot(users, alloc, '-o', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% ----- Graph 2: Utilization (%) -----

subplot(2,1,2)

plot(users, util, '-s', 'LineWidth', 2)

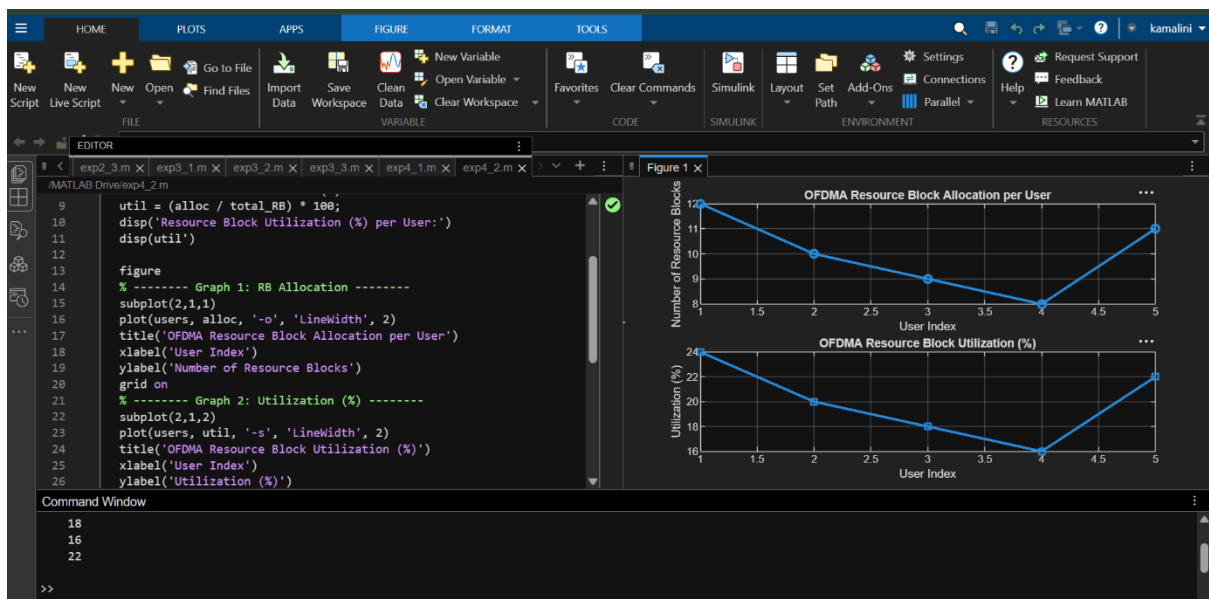
title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on

```



OBJECTIVE 3 :

To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

CODE:

```
clc; clear; close all;

% System parameters

rb_bw = 180e3; % Bandwidth per RB (180 kHz)

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput)

figure

% ----- Graph 1: Throughput per User -----

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')
```

grid on

